



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

Reinforcement Learning (AI3000)

Lecture 05

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The One-Period Problem

- Consider a policy π that stops after taking a single action

$$S_0 \xrightarrow{A_0} (S_1, R_1)$$

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Optimal Value Function

$$Q_1(s, a) := \sum_{s' \in \mathcal{S}} P(s' \mid s, a) r_1(s, a, s')$$

$$v_1^*(s) = \max_{a \in \mathcal{A}} Q_1(s, a)$$

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Optimal Policy

Choose any $a^*(s) \in \arg \max_{a \in \mathcal{A}} Q_1(s, a)$.

$$\pi_1^*(a^*(s) \mid s) = 1$$

Markov, deterministic

Finite Horizon MDPs

- Consider an MDP with a finite horizon N
- Under any $\pi \in \Pi^{\text{MR}}$, we have

$$\begin{aligned} v_{1:N}^{\pi}(s) &= \mathbb{E}^{\pi} \left[\sum_{n=1}^N R_n \mid \{S_0 = s\} \right] = \mathbb{E}^{\pi} [R_1 \mid \{S_0 = s\}] + \mathbb{E}^{\pi} \left[\sum_{n=2}^N R_n \mid \{S_0 = s\} \right] \\ &= \mathbb{E}^{\pi} [R_1 \mid \{S_0 = s\}] + \mathbb{E}^{\pi} \left[\mathbb{E}^{\pi} \left[\sum_{n=2}^N R_n \mid S_1 \right] \mid \{S_0 = s\} \right] = \mathbb{E}^{\pi} [R_1 + v_{2:N}^{\pi}(S_1) \mid \{S_0 = s\}]. \end{aligned}$$

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$$v_{1:N}^{\pi}(s) = \mathbb{E}^{\pi} [R_1 + v_{2:N}^{\pi}(S_1) \mid \{S_0 = s\}], \quad s \in \mathcal{S}.$$

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Backwards Induction

- For each $s \in \mathcal{S}$, let

$$v_{N:N}(s) := \mathbb{E}^{\pi} \left[R_N \mid \{S_{N-1} = s\} \right].$$

- Then, for each $n \in \{1, \dots, N-1\}$, we have

$$v_{n:N}^{\pi}(s) = \mathbb{E}^{\pi} \left[R_n + \sum_{t=n+1}^N R_t \mid \{S_{n-1} = s\} \right] = \mathbb{E}^{\pi} \left[R_n + v_{n+1:N}^{\pi}(S_n) \mid \{S_{n-1} = s\} \right].$$

- The approach above to compute $v_{n:N}$ using $v_{n+1:N}$ is called backwards induction or **dynamic programming (DP)**

Example DP Calculations Under a Markovian Policy

- Suppose that we are given $\pi \in \Pi^{\text{MR}}$
- Then, we have

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- As before, for each $n \in \{1, \dots, N-1\}$, we have

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- Finally, we have

$$v_N^{\pi}(s) = v_{1:N}^{\pi}(s), \quad s \in \mathcal{S}.$$

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DP Calculations under a History-Dependent Policy

- Suppose that we are given $\pi \in \Pi^{\text{HR}}$
- For any $\mathbf{h} = (s_0, a_0, \dots, s_{N-1}) \in \mathcal{H}_{N-1}$, set

$$\begin{aligned} v_{N:N}^{\pi}(\mathbf{h}) &= \mathbb{E}^{\pi} \left[R_N \mid \{H_{N-1} = \mathbf{h}\} \right] \\ &= \sum_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} \pi_{N-1}(a \mid \mathbf{h}) \cdot P(s' \mid s_{N-1}, a) \cdot r_N(s_{N-1}, a, s'). \end{aligned}$$

- Then, for each $n \in \{1, \dots, N-1\}$ and for each $\mathbf{h} = (s_0, a_0, \dots, s_{n-1}) \in \mathcal{H}_{n-1}$, we have the backwards induction

$$v_{n:N}^{\pi}(\mathbf{h}) = \mathbb{E}^{\pi} \left[R_n + \sum_{t=n+1}^N R_t \mid \{H_{n-1} = \mathbf{h}\} \right] = \mathbb{E}^{\pi} \left[R_n + v_{n+1:N}^{\pi}(H_n) \mid \{H_{n-1} = \mathbf{h}\} \right],$$

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Optimal Policies in Finite-Horizon MDPs

- Recall that a policy π^* is optimal in a finite-horizon MDP with horizon N if

$$v_N^{\pi^*}(s) \geq v_N^\pi(s) \quad \forall s \in \mathcal{S}, \quad \forall \pi \in \Pi^{\text{HR}}.$$

Optimal Value Function

$$v_N^*(s) = \sup_{\pi \in \Pi^{\text{HR}}} v_N^\pi(s)$$

Optimal Policy

$$\pi^* \in \operatorname{argsup}_{\pi \in \Pi^{\text{HR}}} v_N^\pi(s)$$

- An optimal policy may not always exist.
- If optimal policies do not exist, we try to find an ε -optimal policy

ε -Optimal Policy

Definition. Given $\varepsilon > 0$, a policy $\pi^\varepsilon \in \Pi^{\text{HR}}$ is said to be ε -optimal if

$$v_N^{\pi^\varepsilon}(s) \geq v_N^*(s) - \varepsilon \quad \forall s \in \mathcal{S}.$$

Of course, this notion of ε -optimality makes sense only when $v_N^*(s)$ is finite for each $s \in \mathcal{S}$.

ϵ -Optimal Policies Always Exist!

- Consider any finite-horizon MDP for which $v_N^*(s)$ is finite for every $s \in \mathcal{S}$
- Fix $\epsilon > 0$, and consider

$$v_N^*(s) = \sup_{\pi \in \Pi^{\text{HR}}} v_N^\pi(s).$$

- By the defining property of the supremum, for every state s , there exists a policy $\pi^{s,\epsilon} \in \Pi^{\text{HR}}$ such that

$$v_N^{\pi^{s,\epsilon}}(s) \geq v_N^*(s) - \epsilon.$$

These statewise policies $\{\pi^{s,\epsilon}\}_{s \in \mathcal{S}}$ may differ from each other
The remaining step is to combine them into a single policy

Combine the Statewise Policies

- Suppose that the MDP starts off in the initial state $S_0 = s_0$
- Recall that the history at any time n is

$$h_n = (s_0, a_0, \dots, a_{n-1}, s_n).$$

- Define $\pi^\varepsilon = \{\pi_n^\varepsilon\}_{n \geq 0}$ by

$$\mathbb{P}^\pi(\{A_n = a\} \mid H_n = h_n) = \pi_n^\varepsilon(a \mid h_n) := \pi_n^{s_0, \varepsilon}(a \mid h_n), \quad n = 0, \dots, N-1.$$

- Clearly, $\pi^\varepsilon \in \Pi^{\text{HR}}$: it simply chooses which statewise policy to follow based on the initial state
- If $S_0 = s$, then π^ε follows $\pi^{s, \varepsilon}$ at every subsequent time step, and therefore

$$v_N^{\pi^\varepsilon}(s) = v_N^{\pi^{s, \varepsilon}}(s) \geq v_N^*(s) - \varepsilon \quad \forall s \in \mathcal{S}.$$

Therefore, for every $\varepsilon > 0$, an ε -optimal policy exists.